Burhan Karahasan

Medical Robotics · Control & Autonomy · Medical Imaging

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EDUCATION

Koç University, Istanbul — M.Sc., Mechanical Engineering *CGPA 4.0/4.0 · 2024 – Present*

Koç University, Istanbul — B.Sc., Mechanical Engineering (Third Rank) *2018 – 2024*

Double Major: B.Sc., Electrical & Electronics Engineering *CGPA 3.95/4.0 · Summa Cum Laude*

RESEARCH EXPERIENCE

X-ray image-guided control of a soft magnetic millirobot (Prof. Metin Sitti's lab) *2024 – Present*

- Built the platform end to end inside a cabinet X-ray system: motorized stages and an actuated end-effector, live image capture from the X-ray unit, calibration, and real-time control software. Now moving the platform to a professional robotic manipulator.
- Designed a finite-state-machine controller that commands the stepper-driven stages in small position steps, chosen over PID because USB latency and stepper actuation make continuous feedback unreliable; it gives stable, precise closed-loop navigation.
- Adapted an existing neural-network detector to locate the robot in low-contrast X-ray frames.
- Video: youtu.be/_P9mQ2cJjDY

Physics-based modeling and simulation of soft magnetic millirobots *2024 – Present*

- Developed physical models of the robot and a beam-based contact simulator: Timoshenko struts, regularized frictional contact, and an implicit solver that refines its steps around contact events.

Ongoing: fabrication-aware inverse design of soft millirobots *2026 – Present*

- Testing the simulator, controller and sensing together on the real robot, to choose robot designs before fabrication.

Koç University Manufacturing & Automation Research Center — Undergraduate Researcher *2021 – 2024*

- Robot-guided surgery: surgical path optimization avoiding critical zones and brain vessels (with Prof. Ismail Lazoglu).
- Mechanical ventilation device: multi-sensor serial communication and system optimization; improved sensor sampling rate ~20×.

PUBLICATIONS & MANUSCRIPTS

[In preparation, first author] B. Karahasan et al. Contact-based physics simulation of untethered magnetic soft millirobots for sensing, control, and inverse design.

[In preparation, first author] B. Karahasan et al. Journal manuscript extending the X-ray navigation work below (title withheld).

[Poster, first author] B. Karahasan, Ö. Yiğit, C. Wang, M. Sitti. “Image-Guided Small-Scale Magnetic Robot Navigation Under X-Ray.” 7th Turkish Robotics Conference, Koç University, İstanbul, July 2026.

[Under review, first author] B. Karahasan, I. Lazoglu, I. Solaroglu. “Entry-point optimization for robot-assisted neurosurgery” (full title withheld pending review). Computer Methods and Programs in Biomedicine.

M. Arshad, E. Guven, B. Karahasan, I. Lazoglu. “A novel real-time wireless sensor integration for enhancing positive pressure system operation in single limb passive vented circuit.” Biomedical Signal Processing and Control 85 (2023).

TECHNICAL SKILLS

Programming: MATLAB, Python, Java/JavaFX, C/C++, VHDL, Visual Basic.

Platforms: Siemens NX, SolidWorks, Fusion 360, Ansys Mechanical, Ansys Fluent.

Areas: Robotics, machine/deep learning, computer vision, finite element analysis, CAD/CAM, numerical methods, generative design, LaTeX.

Languages: English (full professional; all degrees taught in English), Chinese (fundamental).

ACHIEVEMENTS

- Elected Presidential Fellow of Koç University (2024).
- Completed the track program in artificial intelligence.
- Vehbi Koç Honor Award (6 semesters).
- DEVAK Scholarship
- Full merit scholarship for undergraduate studies at Koç University.

INDUSTRY EXPERIENCE

Ubicro — Prototyping & Testing Engineer *1 month - 2022*
DeltaV Space Technologies — GN&C Intern *4 months - 2022*
Turkish Aerospace Industries — Integrated Logistics Support Intern *1 month - 2021*

LEADERSHIP & TEACHING

- Team Leader, Koç University Autonomous Drone Team (KUADRONE), TÜBİTAK UAV Competition (2020–2021): mission planning, thrust calculations, circuit design.
- Peer academic support: Calculus I, Statics, Dynamics, Dynamical Modeling & Control, Senior Design Project.

OBJECTIVE

Robotics researcher with end-to-end experience across robot hardware, imaging, real-time control, and physics-based modeling of soft robots. I aim to pursue a PhD in the control and autonomy of small-scale and medical robots.

REFERENCES

Prof. Metin Sitti — M.Sc. advisor; President, Koç University — msitti@ku.edu.tr

Prof. Ismail Lazoglu — Undergraduate research advisor, Koç University — ilazoglu@ku.edu.tr